

Q.1 Match List-I with List-II

List-I

Differential Equations

(A) $\frac{dy}{dx} + e^y = 0$

(B) $\frac{d^2y}{dx^2} = \left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}$

(C) $\left(\frac{d^2y}{dx^2}\right)^2 + e\left(\frac{dy}{dx}\right) = 0$

(D) $\frac{d^2y}{dx^2} + x\frac{dy}{dx} - 2y = \log x; x > 0$

List-II

Order and degree

(I) order 2, degree not defined

(II) order 2, degree 1

(III) order 1, degree 1

(IV) order 2, degree 2

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

2. (A) - (I), (B) - (IV), (C) - (III), (D) - (II)

3. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

4. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.2 Let the matrix $A = [a_{ij}]_{3 \times 3}$ be defined by

$$a_{ij} = \begin{cases} 2i + 3j, & i < j \\ 5, & i = j \\ 3i - 2j, & i > j \end{cases}$$

The number of elements in the matrix A which are greater than 7, is:

1. 2

2. 3

3. 4

4. 5

Options 1. 1

2. 2

3. 3

4. 4

Q.3

Let A be any square matrix of order 3 and $B = \begin{bmatrix} 0 & -4 & 2 \\ 4 & 0 & 3 \\ -2 & -3 & 0 \end{bmatrix}$.Then the matrix ABA^T is a:

1. Symmetric matrix

2. Skew-symmetric matrix

3. Null matrix

4. Diagonal matrix

Options 1. 1

2. 2

3. 3

4. 4

Q.4 Let the random variable X represent the positive difference between the number of heads and the number of tails obtained when a coin is tossed 6 times. Then probability $P(X \leq 3)$ is equal to.1. $21/64$ 2. $11/32$ 3. $25/32$ 4. $11/64$

Options 1. 1

2. 2

3. 3

4. 4

Q.5 The maximum value of the objective function $Z = 8x + 2y$ of an L.P.P. subject to the constraints: $2x + y \leq 3$, $2x + 3y \leq 6$, $x \geq 0$, $y \geq 0$ is :

1. 12
2. 9
3. 16
4. 6

Options

1. 1
2. 2
3. 3
4. 4

Q.6 $\int_{-1}^1 (x^7 + x^5 + x^3 + x + 1) dx$ is equal to:

1. 0
2. 1
3. 2
4. 4

Options

1. 1
2. 2
3. 3
4. 4

Q.7 The greatest possible value of a such that the function $f(x) = x^2 + ax + 1$ is always decreasing in the interval $[1, 2]$, is:

1. -2
2. -4
3. 2
4. 4

Options

1. 1
2. 2
3. 3
4. 4

Q.8 For $x \in \mathbb{R} - \{-1, 0, 1\}$,

$\int \frac{1}{x - x^5} dx$ is equal to

1. $\frac{1}{4} \log_e \left| \frac{x^4}{1 - x^4} \right| + c$: where c is constant of integration.
2. $-\frac{1}{4} \log_e \left| \frac{x^4}{1 - x^4} \right| + c$: where c is constant of integration.
3. $4 \log_e \left| \frac{x^4}{1 - x^4} \right| + c$: where c is constant of integration.
4. $-4 \log_e \left| \frac{x^4}{1 - x^4} \right| + c$: where c is constant of integration.

Options

1. 1
2. 2
3. 3
4. 4

Q.9 The region represented by the constraints $x \geq 0, y \geq 0$ of an LPP is

1. First quadrant
2. Second quadrant
3. Third quadrant
4. Fourth quadrant

Options

1. 1
2. 2
3. 3
4. 4

Q.10 Particular solution of the differential equation $x(1 + y^2) dx - y(1 + x^2) dy = 0$, given that $y = 0$ when $x = 1$, is:

1. $(1 + x^2) = 2(1 + y^2)$

2. $(1 + y^2) = 2(1 + x^2)$

3. $(1 + x^2) = 4(1 + y^2)$

4. $(1 + y^2) = 4(1 + x^2)$

Options 1. 1

2. 2

3. 3

4. 4

Q.11 Match List-I with List-II

The function $f(x) = 2x^3 - 15x^2 + 36x + 5$ for $x \in [2, 5]$ has

List-I List-II

(A) absolute maximum value (I) 5

(B) absolute minimum value (II) 60

(C) point of absolute maxima (III) 3

(D) point of absolute minima (IV) 32

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

2. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

3. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)

4. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

Options 1. 1

2. 2

3. 3

4.

4

Q.12

If $A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 2 \\ 2 & -1 & \lambda \end{bmatrix}$ is a singular matrix, then the value of λ is:

1. 0

2. 3

3. 5

4. -5

Options 1. 1

2. 2

3. 3

4. 4

Q.13

The area of the region (in square units) bounded by $x = 1$, $x = 2$ and the curve $y^2 = 4x$ in the first quadrant is:

1. $\frac{4}{3}[2\sqrt{2} - 1]$

2. $\frac{2}{3}[2\sqrt{2} - 1]$

3. $\frac{2}{3}[2\sqrt{2} + 1]$

4. $\frac{4}{3}[2\sqrt{2} + 1]$

Options 1. 1

2. 2

3. 3

4. 4

Q.14

If $y = \frac{1}{1 + x^{b-a} + x^{c-a}} + \frac{1}{1 + x^{c-b} + x^{a-b}} + \frac{1}{1 + x^{a-c} + x^{b-c}}$ then $\frac{d^2y}{dx^2}$ is:

1. $x^a + x^b + x^c$

2. $(x^a + x^b + x^c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right)$

3. 0

4. 1

Options 1. 1

2. 2

3. 3

4. 4

Q.15

If $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \\ x & 1 & 1 \end{bmatrix}$ and $A^{-1} = \frac{1}{4} \begin{bmatrix} -2 & 0 & y \\ 5 & -2 & -1 \\ 1 & 2 & -1 \end{bmatrix}$, then values of x and y are:

1. $x = 3, y = 2$

2. $x = 2, y = 3$
3. $x = -3, y = -2$
4. $x = 3, y = -2$

Options 1. 1
2. 2
3. 3
4. 4

Section : Applied Mathematics

Q.16 Which of the following are NOT correct about "Sinking Fund"?

- (A) It does not have any specific purpose.
- (B) It can be used in any emergency.
- (C) Any amount, any time can be deposited in it.
- (D) It is set up for a particular upcoming expense.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (C) and (D) only
4. (B), (C) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

Q.17 The total cost $C(x)$ associated with the production of x units of an item is given by

$$C(x) = 0.001x^3 + 0.06x^2 + 20x + 500$$

The marginal cost when 10 units are produced is:

1. 21.5
2. 500
3. 31.5
4. 27.5

Options 1. 1
2. 2
3. 3
4. 4

Q.18 If the objective function $Z = px + qy$, $p > 0$, $q > 0$ of a linear programming problem attains its optimal value at the points (4, 7) and (5, 5) and $pq = 50$ then

1. $p = 5$, $q = 10$
2. $p = 10$, $q = 5$
3. $p = 25$, $q = 2$
4. $p = 40/3$, $q = 15/4$

Options 1. 1
2. 2
3. 3
4. 4

Q.19 Which of the following are NOT correct regarding the equation of tangent and normal to the curve $y = \frac{x-11}{(x-2)(x-3)}$ at the point, where it cuts the x-axis?

- (A) The point of contact is (11, 0).
- (B) The equation of tangent is $x - 72y - 11 = 0$.
- (C) The equation of normal is $72x + y - 11 = 0$.
- (D) The slope of the tangent at the given point of contact is $\frac{1}{88}$.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (B) and (C) only
3. (A), (C) and (D) only
4. (C) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

Q.20 Which of the following inequalities are NOT correct?

- (A) If $a > 1, b > 1$, then $\log_a a + \log_a b \leq 2$
- (B) For any real number x , $(9^x + 9^{1-x}) \geq 9$
- (C) If a, b, c are non-zero real numbers of the same sign, then $\left(\frac{a}{b} + \frac{b}{c} + \frac{c}{a}\right) \leq 3$
- (D) If a, b, c are three distinct real numbers, then $(a+b)(b+c)(c+a) \geq 8abc$

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (C) and (D) only
4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.21 The normal distribution curve is symmetrical about
[μ = mean, σ = standard deviation]

1. $x = \mu^{\frac{1}{2}}$

2. $x = \mu$

3. $x = \sigma$

4. $x = \frac{\mu}{\sigma}$

Options 1. 1

2. 2

3. 3

4. 4

Q.22 A machine costing Rs. 25000 has a useful life of 4 years. The estimated scrap value is Rs. 5000. The annual depreciation by linear method is:

1. Rs. 6000

2. Rs. 5000

3. Rs. 5500

4. Rs. 4500

Options 1. 1

2. 2

3. 3

4. 4

Q.23 A person has an initial investment of 25000 in an investment plan. After 2, years it has grown 30000, then the rate of return on his investment is

1. 10%

2. 25%

3. 20%

4. 15%

Options 1. 1

2. 2

3. 3

4. 4

Q.24 A specific characteristic of a sample is known as a

1. Population

2. Parameter

3. Population distribution

4. Statistic

Options 1. 1

2. 2

3. 3

4. 4

Q.25 If A is a square matrix of order 3 such that $|A| = 3$ then $|\text{adj}(A)|$ is equal to:

1. $\frac{1}{3}$

2. 3

3. 6

4. 81

Options 1. 1

2. 2

3. 3

4. 4

Q.26 The general solution of the differential equation $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$ is equal to:

1. $e^y = e^x + \frac{x^3}{3} + c$, where c is an arbitrary constant.

2. $e^y = x e^x + \frac{x^2}{3} + c$, where c is an arbitrary constant.

3. $e^y = 2e^x + \frac{x^2}{4} + c$, where c is an arbitrary constant.

4. $y = e^x + \frac{x^3}{3} + c$, where c is an arbitrary constant.

Options 1. 1

2. 2

3. 3

4. 4

Q.27 Due to which of the following, the irregular variations in a time series are caused:

(A) Floods

(B) Rise in prices before festivals

(C) A fire in a factory

(D) Epidemics

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (C) and (D) only
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.28

If the matrix $\begin{bmatrix} 0 & 1 & 4x \\ -1 & 0 & -5 \\ 2 & 5 & y \end{bmatrix}$ is skew-symmetric, then

1. $x = -1, y = 0$
2. $x = -\frac{1}{2}, y = 0$
3. $x = 1, y = -1$
4. $x = 0, y = -1$

Options 1. 1

2. 2
3. 3
4. 4

Q.29

In what ratio should a shopkeeper mix two types of rice, one costing ₹20 per kg and another costing ₹40 per kg to get a rice variety costing ₹28 per kg?

1. 3 : 5
2. 3 : 2
3. 3 : 4
4. 4 : 3

Options 1. 1

2. 2
3. 3
4. 4

Q.30

In a game, a person is paid Rs. 2 if he gets all heads or all tails when three coins are tossed and he will pay Rs. 2 if either one or two heads show. What can he expect to win on an average per game?

1. Lose Rs. 1 per toss of the coins.
2. Win Rs. 1 per toss of the coins.
3. Lose Rs. 2 per toss of the coins.
4. No profit, no loss.

Options 1. 1

2. 2
3. 3
4. 4

Q.31 The present value of a sequence of payments of Rs. 2000 made at the end of every 6 months and continuing forever, if money is worth 8% per annum compounded semi-annually, is:

1. Rs. 25000
2. Rs. 50000
3. Rs. 40000
4. Rs. 5000

Options 1. 1

2. 2
3. 3
4. 4

Q.32

A lot of 50 watches is known to have 10 defective watches. If 8 watches are selected one by one with replacement at random, then the probability that there will be at least one defective watch is:

1. $1 - \left(\frac{4}{5}\right)^8$
2. $1 - \left(\frac{4}{5}\right)^{10}$
3. $1 - \left(\frac{1}{5}\right)^2 \left(\frac{4}{5}\right)^8$
4. $1 - \left(\frac{1}{5}\right)^2 \left(\frac{4}{5}\right)^6$

Options 1. 1

2. 2

3. 3
4. 4

Q.33 A person can row a boat at 5 km/hr in still water. If the speed of water current in a river is 1 km/hr, and it takes him 1 hour to row to a place and come back, how far off is the place?

1. 3.6 km
2. 2.4 km
3. 2 km
4. 3.2 km

Options 1. 1

2. 2
3. 3
4. 4

Q.34 For the given 5 values 24, 18, 33, 42, 24; the 3-year moving averages are:

1. 19, 22, 33
2. 25, 31, 33
3. 25, 29, 31
4. 22, 29, 31

Options 1. 1

2. 2
3. 3
4. 4

Q.35 Mr. X wishes to purchase a house for ₹ 14,51,400 from a bank and decided to repay the loan by equal monthly installments (EMI) in 10 years. If bank charges interest at 9% per annum compounded monthly, then the EMI is:

[Given that $(1.0075)^{120} = 2.4514$]

1. ₹ 18385.50
2. ₹ 15627.42
3. ₹ 10582.42
4. ₹ 18564.50

Options 1. 1

2. 2
3. 3
4. 4

Q.36

If $A = \begin{bmatrix} 4 & 5 \\ 2 & 1 \end{bmatrix}$ and I is an identity matrix of order 2, then $A - 3I$ equals

1. $I + 3A^2$
2. $I + 3A^{-1}$
3. $2(3I + A^{-1})$
4. $2(I + 3A^{-1})$

Options 1. 1

2. 2
3. 3
4. 4

Q.37

If $\int \frac{dx}{(x-1)^{3/4}(x+2)^{5/4}} = \alpha[1 - g(x)]^\beta + c$, where c is a constant of integration, then which of the following are true?

(A) $\alpha = \frac{2}{3}$

(B) $\beta = \frac{3}{4}$

(C) $3\alpha + 4\beta = 5$

(D) $g(x) = \frac{3}{x+2}$

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.38 Three pipes A, B and C can fill a tank in 12 hours, 15 hours and 20 hours respectively. If A is open all the time and B and C are open for one hour each alternately, in how many hours will the tank be full?

1. 6 hours
2. 5.5 hours
3. 8 hours
4. 7 hours

Options 1. 1

2. 2
3. 3

Q.39

The number of all possible matrices of order 2×3 with entries -1 or 1 is

1. 64
2. 6
3. 32
4. 16

Options 1. 1

2. 2
3. 3
4. 4

Q.40

Consider the linear programming problem (LPP):

Minimize $Z = x + y$

subject to

$$x + 2y \leq 4,$$

$$3x + y \geq 3,$$

$$4x + 3y \geq 6,$$

$$x, y \geq 0.$$

Which of the following is correct for the above linear programming problem (LPP):

- (A) The LPP has a bounded feasible region.
- (B) The LPP has a unique optimal solution.
- (C) The optimal value of the LPP exists at the point $(3/2, 0)$
- (D) The corner points of the feasible region are $(3/2, 0)$, $(3/5, 6/5)$, $(2/5, 6/5)$ and $(4, 0)$

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.41 Which of the following are the components of a time series?

- (A) Cyclic component
- (B) Regular component
- (C) Seasonal component
- (D) Economic component

Choose the correct answer from the options given below:

1. (A) and (C) only
2. (B) and (C) only
3. (A) and (B) only
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.42 If we take 8 identical slips of paper and write the number 0 on one of them, the number 1 on three of the slips, the number 2 on three of the slips and the number 3 on one of the slips. These slips are folded, put in a box and roughly mixed. One slip is drawn at random from the box. If X is the random variable denoting the number written on the drawn slip, the variance of X is:

1. $1/4$
2. $3/4$
3. 2
4. $3/2$

Options 1. 1

2. 2
3. 3
4. 4

Q.43 For the system of equations $AX = B$, which of the following is correct?

1. If A is a singular matrix, then $AX = B$ has a unique solution.
2. If A is a non-singular matrix, then $AX = B$ has infinitely many solutions.
3. If A is a non-singular matrix, then $AX = B$ has a unique solution.
4. If A is a singular matrix, then $AX = B$ is always consistent.

Options 1. 1

2. 2
3. 3
4. 4

Q.44 A runs 3 times as fast as B. If A gives B a start of 30 meters, how far must the goal on the race course be so that A and B reach it at the same time?

1. 80 m
2. 42 m
3. 45 m
4. 54 m

Options 1. 1

2. 2
3. 3
4. 4

Q.45

If an investment of Rs. 12000 becomes Rs. 72000 in 4 years, then the compound annual growth rate is:

1. $\frac{\sqrt[4]{6} - 1}{100}\%$
2. $\frac{\sqrt[4]{6} + 1}{100}\%$
3. $(\sqrt[4]{6} + 1) \times 100\%$
4. $(\sqrt[4]{6} - 1) \times 100\%$

Options 1. 1

2. 2
3. 3
4. 4

Q.46

The value of the definite integral $I = \int_1^2 \frac{1}{x(1+x^2)} dx$ is:

1. $\frac{3}{2}$
2. $\frac{3}{2} \log 2 - \frac{1}{2} \log 5$
3. $\frac{5}{2} \log 3 - \frac{1}{2} \log 5$
4. $\frac{5}{2} \log 3 + \log 5$

Options 1. 1

2. 2
3. 3
4. 4

Q.47

The function $f(x) = kx^3 + 6kx^2 + 18x + 17$ is increasing on $\mathbb{R}$ (set of real numbers) if:

1. $k \in (0, 3)$
2. $k \in (0, \frac{3}{2})$
3. $k \in (\frac{1}{2}, \frac{3}{2})$
4. $k \in (\frac{1}{2}, \frac{5}{2})$

Options 1. 1

2. 2
3. 3
4. 4

Q.48

If $x^2 - y^2 = t - \frac{1}{t}$, and $x^4 + y^4 = t^2 + \frac{1}{t^2}$, then which of the following is correct?

1. $xy^3 \frac{dy}{dx} + 1 = 0$
2. $x^3 \frac{dy}{dx} + y = 1$
3. $x^2 y \frac{dy}{dx} - 1 = 0$
4. $x^3 y \frac{dy}{dx} + 1 = 0$

Options 1. 1

2. 2
3. 3
4. 4

Q.49

The least non-negative remainder when 2^{101} is divided by 5 is:

1. 1
2. 5
3. 2
4. 4

Options 1. 1

2. 2
3. 3
4. 4

Q.50 The simple random sample consists of six observations: 5, 8, 10, 7, 10, 14. The point estimate of the population mean is:

1. 1
2. 4
3. 9
4. 25

- Options 1. 1
2. 2
3. 3
4. 4

Section : Core Mathematics

Q.51 Consider the region bounded by the lines $y - 1 = x$, $x = -2$, $x = 3$ and x - axis. Then

(A) The area of the bounded region is given by $\int_{-2}^3 (x + 1) dx$

(B) The numerical value of the area is $\frac{15}{2}$ sq. units

(C) The numerical value of the area is 8 sq. units

(D) The numerical value of the area is $\frac{17}{2}$ sq. units

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (A) and (C) only
3. (C) and (D) only
4. (D) only

- Options 1. 1
2. 2
3. 3
4. 4

Q.52 Let $\vec{a} = \hat{i} + \hat{j}$, $\vec{b} = \hat{i} - \hat{j}$ and $\vec{c} = \hat{i} + \hat{j} + \hat{k}$. If $\hat{m}$ is a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$, then $|\vec{c} \cdot \hat{m}|$ is equal to

1. 4
2. 2
3. 0
4. 1

- Options 1. 1
2. 2
3. 3
4. 4

Q.53 Match List-I with List-II

List-I	List-II
Lines	Direction Ratios
(A) $\frac{x-1}{2} = \frac{2-y}{1} = z$	(I) (1, 3, -1)
(B) $\frac{2x-1}{2} = \frac{y+1}{3} = \frac{1-z}{1}$	(II) (2, -2, 0)
(C) $\frac{x+1}{2} = \frac{3-y}{2} = z = 2$	(III) (2, -1, 1)
(D) $\frac{2x-3}{4} = \frac{1-2y}{2} = \frac{z}{5}$	(IV) (2, -1, 5)

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
2. (A) - (III), (B) - (I), (C) - (II), (D) - (IV)
3. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)
4. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)

- Options 1. 1
2. 2
3. 3
4. 4

Q.54 The angle between the lines

$$l_1 : \frac{x+1}{1} = \frac{2-y}{2} = \frac{z-1}{1} \text{ and } l_2 : \frac{x-1}{4} = \frac{2y-4}{6} = \frac{z-1}{2}, \text{ is}$$

1. $\frac{\pi}{3}$

2. $\frac{\pi}{4}$
3. $\frac{\pi}{6}$
4. $\frac{\pi}{2}$

Options 1. 1

2. 2
3. 3
4. 4

Q.55 If $\begin{vmatrix} x & 8 \\ 4 & x \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix}$, then x is/are equal to

1. $\pm 4\sqrt{2}$
2. $2\sqrt{2}$
3. $4\sqrt{3}$
4. $\pm 4\sqrt{3}$

Options 1. 1

2. 2
3. 3
4. 4

Q.56 Match List-I with List-II

Consider two vectors $\vec{a} = \hat{i} + 2\hat{j} - \hat{k}$ and $\vec{b} = -3\hat{i} - 6\hat{j} + 3\hat{k}$.

List-I

List-II

- (A) Angle between $\vec{a}$ and $\vec{b}$ (I) $\cos^{-1}\left(\frac{1}{\sqrt{6}}\right)$
- (B) Angle between $\vec{a}$ and x-axis (II) $\cos^{-1}\left(\frac{2}{\sqrt{6}}\right)$
- (C) Angle between $\vec{b}$ and x-axis (III) π
- (D) Angle between $\vec{a}$ and y-axis (IV) $\cos^{-1}\left(-\frac{1}{\sqrt{6}}\right)$

Choose the correct answer from the options given below:

1. (A) – (III), (B) – (IV), (C) – (I), (D) – (II)
2. (A) – (III), (B) – (I), (C) – (IV), (D) – (II)
3. (A) – (III), (B) – (II), (C) – (I), (D) – (IV)
4. (A) – (II), (B) – (III), (C) – (I), (D) – (IV)

Options 1. 1

2. 2
3. 3
4. 4

Q.57 If a person A speaks the truth in 80% cases and the person B speaks the truth in 75% cases, then the probability that they contradict each other in a statement is

1. $\frac{2}{5}$
2. $\frac{13}{20}$
3. $\frac{3}{5}$
4. $\frac{7}{20}$

Options 1. 1

2. 2
3. 3
4. 4

Q.58 The function $f(x) = \frac{x-2}{x+1}$, $x \neq -1$ is increasing when (Where $\mathbb{R}$ is a set of real numbers)

1. $x \in \mathbb{R}$
2. $x \in \mathbb{R} - \{-1\}$
3. $x \in \mathbb{R} - \{1, -1\}$
4. $x \in \mathbb{R} - \{0\}$

Options 1. 1

2. 2

3. 3
4. 4

Q.59 For $x \in [-1, 1]$, if $4 \sin^{-1} x + \cos^{-1} x = \pi$ then x is equal to

1. $-\frac{1}{2}$
2. $\frac{1}{2}$
3. $\frac{\sqrt{3}}{2}$
4. $-\frac{\sqrt{3}}{2}$

Options 1. 1
2. 2
3. 3
4. 4

Q.60 Consider the function $f(x) = \sin x$ in the interval $[\pi, 2\pi]$, then which of the following statements are correct?

- (A) $x = \frac{3\pi}{2}$ is its stationary point.
(B) Its maximum value is 1
(C) Its minimum value is -1
(D) It attains its maximum value at π and 2π

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A) and (C) only
3. (A), (C) and (D) only
4. (B), (C) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

Q.61 A problem in Mathematics is given to two students X and Y whose chances of solving it are $\frac{1}{3}$ and $\frac{1}{4}$ respectively. The probability that only X solves the problem, is:

1. $\frac{1}{3}$
2. $\frac{1}{12}$
3. $\frac{1}{4}$
4. $\frac{7}{12}$

Options 1. 1
2. 2
3. 3
4. 4

Q.62 The values of λ for which the system of equation $x + 2y + z = 14, -x + y + z = 10, x + \lambda y + z = 2$ has unique solution is

1. $\mathbb{R} - \{2\}$, where $\mathbb{R}$ is set of real numbers.
2. $-2 \leq \lambda \leq 2$
3. $-4 < \lambda < 4$
4. 2

Options 1. 1
2. 2
3. 3
4. 4

Q.63 The function $f: [0, \infty) \rightarrow \mathbb{R}$ defined by, $f(x) = 2x^2 + 3$, is

1. one-one and onto
2. one-one but not onto
3. onto but not one-one
4. neither one-one nor onto

Options 1. 1
2. 2
3. 3
4. 4

Q.64 The radius of spherical balloon is decreasing at the rate of 0.1 cm/sec, the rate at which its volume is decreasing, when its radius is 0.5 cm is

1. $\frac{\pi}{100} \text{ cm}^3/\text{sec}$
2. $\frac{\pi}{10} \text{ cm}^3/\text{sec}$
3. $\pi \text{ cm}^3/\text{sec}$
4. $\frac{2\pi}{5} \text{ cm}^3/\text{sec}$

Options 1. 1

2. 2
3. 3
4. 4

Q.65 The minimum value of the objective function $Z = x + 2y$ of an L.P.P. subject to constraints

$$2x + y \geq 3, \frac{x}{2} + 2y \geq 6, x \geq 0, y \geq 0 \text{ is}$$

1. 12
2. 6
3. $\frac{3}{2}$
4. 0

Options 1. 1

2. 2
3. 3
4. 4

Q.66 If A is a square matrix of order 3×3 and $|A| = 4$. The value of $|(\text{adj}A) \cdot A|$ is

1. 4
2. 16
3. 64
4. 128

Options 1. 1

2. 2
3. 3
4. 4

Q.67 If A and B are two invertible matrices, then which of the following statements are correct?

- (A) $|A^{-1}| = |A|^{-1}$
- (B) $\text{adj}A = |A|A^{-1}$
- (C) $(AB)^{-1} = A^{-1}B^{-1}$
- (D) $(A+B)^{-1} = A^{-1} + B^{-1}$

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (A) and (B) only
3. (A), (B) and (D) only
4. (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.68 If $I_n = \int_0^{\pi/4} \tan^n x \, dx$ then $I_{2024} + I_{2026}$ is equal to :

1. $\frac{1}{2025}$
2. $\frac{1}{2027}$
3. $\frac{1}{2023}$
4. $\frac{2}{2025}$

Options 1. 1

2. 2
3. 3
4. 4

Q.69 Area of the region bounded by the curve $y^2 = 4x$, y -axis and the line $y = 3$ is equal to

1. $\frac{9}{2}$ sq. units
2. 9 sq. units
3. 2 sq. units
4. $\frac{9}{4}$ sq. units

Options 1. 1

2. 2
3. 3

Q.70 An urn I contains 3 white and 4 blue balls, while urn II contains 5 white and 6 blue balls. One ball is drawn at random from one of the urns and it is found to be white. The probability that it was drawn from urn II is

1. $\frac{53}{68}$
2. $\frac{5}{38}$
3. $\frac{3}{68}$
4. $\frac{35}{68}$

Options 1. 1

2. 2
3. 3
4. 4

Q.71 $\int \frac{e^x(1+x) dx}{\cos^2(e^x)}$ is equal to

1. $\tan(e^x) + c$, where c is constant of integration .
2. $-\cot(e^x) + c$, where c is constant of integration .
3. $\cot(e^x) + c$, where c is constant of integration .
4. $\tan(xe^x) + c$, where c is constant of integration .

Options 1. 1

2. 2
3. 3
4. 4

Q.72 Let a relation $R = \{(a, b) : a \text{ is a factor of } b, a, b \in \mathbb{N}\}$. Then, R is _____.

1. Reflexive and Symmetric but NOT Transitive.
2. Symmetric and Transitive but NOT Reflexive.
3. Reflexive and Transitive but NOT Symmetric.
4. Equivalence Relation.

Options 1. 1

2. 2
3. 3
4. 4

Q.73 The maximum value of the objective function $Z = 2x + y$ of an LPP, subject to the constraints $x \leq 6, y \leq 2, x - y \leq 0, x \geq 0, y \geq 0$, is

1. 14
2. 6
3. 12
4. 8

Options 1. 1

2. 2
3. 3
4. 4

Q.74 If $\hat{a}$ is a unit vector perpendicular to both the vectors $\vec{b} = \hat{j} + 2\hat{k}$ and $\vec{c} = \hat{i} + 2\hat{j}$, then $\hat{a}$ is equal to

1. $\frac{-4\hat{i} + 2\hat{j} + \hat{k}}{\sqrt{21}}$
2. $\frac{4\hat{i} + 2\hat{j} - \hat{k}}{\sqrt{21}}$
3. $\frac{-4\hat{i} + 2\hat{j} - \hat{k}}{\sqrt{21}}$
4. $\frac{-4\hat{i} - 2\hat{j} - \hat{k}}{\sqrt{21}}$

Options 1. 1

2. 2
3. 3
4. 4

Q.75 If $\vec{a} = 2\hat{i} - 3\hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$, then which of the following statements is/are correct?

- (A) $\vec{a}$ and $\vec{b}$ are collinear
- (B) $\vec{a}$ and $\vec{b}$ are perpendicular
- (C) Angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{4}$
- (D) $|\vec{a} + \vec{b}| = 2\sqrt{5}$

Choose the correct answer from the options given below:

1. (A) and (C) only
2. (B) only
3. (A), (C) and (D) only
4. (B) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

Q.76

General solution of the differential equation $\frac{dy}{dx} = e^{\frac{x^2}{2}} + xy$ is

1. $y = ce^{\frac{x^2}{2}}$, where c is constant of integration .
2. $y = ce^{-\frac{x^2}{2}}$, where c is constant of integration .
3. $y = (x + c)e^{\frac{x^2}{2}}$, where c is constant of integration .
4. $y = (x + c)e^{-\frac{x^2}{2}}$, where c is constant of integration .

Options

1. 1
2. 2
3. 3
4. 4

Q.77

Consider a line $\vec{r} = (\hat{i} + 4\hat{j}) + \lambda(2\hat{i} - 2\hat{j} + 3\hat{k})$, then which of the following statements are correct?

- (A) It passes through point $(9, -4, 12)$
- (B) It passes through point $(1, 4, -1)$
- (C) Its direction cosine's are $\frac{2}{\sqrt{17}}, \frac{-2}{\sqrt{17}}, \frac{3}{\sqrt{17}}$
- (D) Its Cartesian equation is $\frac{x-1}{2} = \frac{y-4}{-2} = \frac{z}{3}$

Choose the correct answer from the options given below:

1. (A), (C) and (D) only
2. (B) and (C) only
3. (A) and (C) only
4. (B) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

Q.78 Match List-I with List-II

List-(I)

List-(II)

Function

Derivative

(A) $y = \sin^{-1} x + \sin^{-1} \sqrt{1-x^2}$, $|x| < 1$ (I) $\frac{dy}{dx} = \frac{1}{2y-1}$

(B) $y = \sqrt{x+y}$, $x+y > 0$, $y \neq \frac{1}{2}$ (II) $\frac{dy}{dx} = 10^x \log_e 10$

(C) $y = \log_{10} x$, $x > 0$ (III) $\frac{dy}{dx} = 0$

(D) $y = 10^x$ (IV) $\frac{dy}{dx} = \frac{1}{x \log_e 10}$

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)
4. (A) - (II), (B) - (I), (C) - (III), (D) - (IV)

Options

1. 1
2. 2
3. 3
4. 4

Q.79

If $A = \begin{bmatrix} x & -3 & 4 \\ 3 & y & -5 \\ -4 & z & 0 \end{bmatrix}$ is a Skew-Symmetric matrix and $\text{adj } A = [a_{ij}]_{3 \times 3}$, then $a_{11} + a_{22} + a_{33}$ is equal to

1. 32

2. 0
3. 50
4. 18

Options 1. 1
2. 2
3. 3
4. 4

Q.80 Match List-I with List-II

List-I

List-II

- (A) $f(x) = [x]$ (I) Continuous everywhere but not differentiable at $x = -1$
(B) $f(x) = |x - 1|$ (II) Continuous everywhere except at all integral values
(C) $f(x) = e^{|x|}$ (III) Continuous everywhere but not differentiable at $x = 1$
(D) $f(x) = |x + 1|$ (IV) Continuous everywhere but not differentiable at $x = 0$

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
3. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
4. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)

Options 1. 1
2. 2
3. 3
4. 4

Q.81 For the differential equation $(x + y)dy + (x - y)dx = 0$, which of the following is/are correct?

- (A) Differential equation is homogeneous
(B) Order of differential equation is 1
(C) Integrating factor of differential equation is e^x
(D) Degree of the equation is not defined

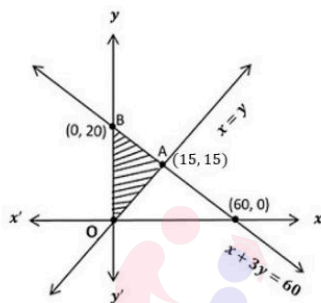
Choose the correct answer from the options given below:

1. (A) only
2. (B), (C) and (D) only
3. (A), (B) and (D) only
4. (A) and (B) only

Options 1. 1
2. 2
3. 3
4. 4

Q.82

For an LPP: Maximize $z = 3x + 9y$; $x \geq 0, y \geq 0$, the feasible region OAB is shown in the figure, then the other constraints are



1. $x + 3y \leq 60, x \geq y$
2. $x + 3y \geq 60, x \leq y$
3. $x + 3y \geq 60, x \geq y$
4. $x + 3y \leq 60, x \leq y$

Options 1. 1
2. 2
3. 3
4. 4

Q.83 $\int_{-\pi}^{\pi} \frac{e^{\sin x}}{e^{\sin x} + e^{-\sin x}} dx$ is equal to

1. 0

2. $\frac{\pi}{2}$
3. π
4. $\frac{\pi}{4}$

Options 1. 1

2. 2
3. 3
4. 4

Q.84

If $y = x \sin y$, then $\frac{dy}{dx}$ is:

1. $\frac{\sin y}{x - \cos y}$
2. $\frac{\sin^2 y}{\sin y - y \cos y}$
3. $\frac{\sin^2 y}{\cos y - y \sin y}$
4. $\frac{\sin y}{x \cos y + y}$

Options 1. 1

2. 2
3. 3
4. 4

Q.85

If $A = \begin{bmatrix} 2 & -3 & 4 \\ -3 & 5 & x \\ 4 & 3 & 0 \end{bmatrix}$ is a symmetric matrix and $B = \begin{bmatrix} 0 & 2 & -10 \\ -2 & z & 6 \\ y & -6 & 0 \end{bmatrix}$ is a skew-symmetric matrix, then the value of

$(xy + yz + zx)$, is

1. 0
2. 30
3. -30
4. 60

Options 1. 1

2. 2
3. 3
4. 4